

# Pavani Jain

[jain.pav@northeastern.edu](mailto:jain.pav@northeastern.edu) | [github.com/Pavanijain03](https://github.com/Pavanijain03) | [linkedin.com/in/pavani-jain](https://www.linkedin.com/in/pavani-jain) | [pavanijain03.github.io](https://pavanijain03.github.io) |  
(617)-516-9584

## Education

### Northeastern University

*Master of Science in Artificial Intelligence* | GPA: 3.41/4.00

Expected May 2027

Boston, MA

### Guru Gobind Singh Indraprastha University

*Bachelor of Computer Applications (BCA)* | GPA: 3.64/4.00

Aug 2022 – Jun 2025

Delhi, India

## Skills

**Languages:** Python, R, Java, C, C++, JavaScript, Bash, SQL, HTML, CSS

**ML/AI:** Deep Learning, NLP, Generative AI, LLMs, RAG, Agentic AI, AI Orchestration, MLOps, Anomaly Detection, Prompt Engineering, A/B Testing

**Libraries:** PyTorch, TensorFlow, scikit-learn, Pandas, NumPy, LangChain, LangGraph, HuggingFace, Transformers, OpenAI API, Anthropic API, SpaCy, PySpark

**Data & Tools:** PostgreSQL, pgvector, FAISS, MongoDB, BigQuery, Snowflake, ETL, Tableau, Power BI, FastAPI, AWS (SageMaker, S3, EC2, Bedrock), Docker, Kubernetes, Git, MLflow, CI/CD

## Experience

### AI Engineer Intern

May 2026 – Present

*Thingqr* Austin, TX

- Architected the AI orchestration layer (Python, LangChain, LangGraph, OpenAI/Anthropic APIs) for 5+ multi-agent, multi-turn interview workflows with stateful graphs, 8+ tool-calling nodes, and structured output across 3 seniority tiers and 4 role families; built an LLM-as-judge eval pipeline scoring 6 competency dimensions over 50+ behavioral signals with guardrails, hallucination checks, and chain-of-thought traces for auditable scorecards.
- Developed a production RAG pipeline (FAISS, pgvector, sentence-transformers, hybrid retrieval) over 100+ scenarios at ~90% top-5 accuracy, deployed on Dockerized AWS (EC2, Bedrock) workspaces with FastAPI microservices – cutting spin-up from ~2 min to ~25 sec and P95 LLM latency to ~1.8 sec.

### ML Researcher – Northeastern University

Oct 2025 – Dec 2025

*Anomaly Detection in Financial Transactions Using Deep Learning*

Boston, MA

- Authored a research paper on autoencoder-based fraud detection on imbalanced data (<0.2% fraud); reviewed 25+ works and benchmarked 4 architectures (LR, RF, XGBoost, Autoencoder) with SMOTE, undersampling, feature engineering, threshold tuning, and ablation studies, reporting ROC-AUC, precision-recall, and F1.

### Data Analyst Intern

Jun 2024 – Sep 2024

*Verve Bridge*

Delhi, India

- Identified a ~6% payment success-rate gap via 50+ SQL queries (CTEs, window functions, joins) on 50K+ transactions, slicing KPIs by payment method, bank, and merchant – surfacing 3 underperforming routes that triggered an A/B test on retry logic.
- Built an end-to-end ETL pipeline in Python (Pandas, NumPy, SQLAlchemy) consolidating logs, settlements, and chargebacks from PostgreSQL – cutting manual prep from ~3 hrs to ~20 mins with data-quality checks flagging 10+ weekly exceptions; shipped 3 Tableau dashboards adopted in weekly business reviews, enabling self-serve analytics for product, ops, and growth.

## Projects

### OCT Retinal Disease Detection – DenseNet Transfer Learning

May 2026 – Jun 2026

*Python, PyTorch, DenseNet-121, Transfer Learning, Streamlit*

- Fine-tuned a DenseNet-121 transfer-learning classifier (frozen ImageNet backbone, retrained head) on 100K+ labeled OCT retinal scans to flag CNV, DME, and drusen, reaching 95.4% test accuracy, 96.7% recall, and 98.8% ROC-AUC on a held-out set.
- Countered class imbalance with a weighted cross-entropy objective and a standardized preprocessing pipeline (grayscale-to-RGB, 224×224 center-crop, ImageNet normalization); benchmarked ResNet, VGG, and DenseNet backbones and shipped the top model as a Streamlit inference app.

### Nanochat – GPT Pretraining & LLM Systems Optimization

Mar 2026 – Apr 2026

*Python, PyTorch, Transformers*

- Built a GPT-style LM from scratch: BPE tokenizer, multi-head attention, training loop, and chat interface; trained on FineWeb-Edu with 8×H100 GPUs, reaching GPT-2-level validation loss in ~3 hrs at ~\$72 compute cost.
- Studied compute-quality tradeoffs by varying transformer depth, learning rate schedules, and batch sizing.

### Genomic Text Curation & Topic Grouping

Jan 2026 – Feb 2026

*Python, NLP, Scikit-learn, TF-IDF, Topic Modeling*

- Built an NLP pipeline extracting genes, variants, and diseases from 50 genomics abstracts; produced 76 knowledge-graph-ready triples (40 variants, 37 genes, 19 diseases) with 84% variant and 78% gene recall.
- Compared TF-IDF + K-Means, NMF, and LDA for topic modeling; identified 5 coherent biomedical clusters.

## Activities

### Participant, NU Hacks (48-hour hackathon)

March 2026

*Northeastern University*

Boston, MA

- Built a RAG clinical Q&A prototype with LangChain, indexing 1,000+ medical docs for cardio/oncology queries; trained an LSTM on 500+ simulated patient timelines to flag early heart and cancer risk indicators.
- Optimized retrieval via ANN search and prompt tuning, achieving sub-2-second average query response time.